CoolGrid — The AI-Powered Thermal Intelligence Platform for Data Centers

> *"Every degree of cooling efficiency saved is $1 million recovered. We see the heat before it becomes a crisis."*

The $50 Billion Cooling Crisis

The world's data centers are overheating — and today's tools can't keep up.

In May 2026, AWS suffered a major outage when a Virginia data center overheated, disrupting services for millions. This isn't an anomaly — it's the new normal:

- **AI workloads are 10x more power-intensive** than traditional compute — GPUs generate 700W+ per chip
- **Data center energy consumption:** 4% of global electricity and growing 15% annually
- **Cooling accounts for 40%** of total data center energy costs
- **Stranded capacity:** $200B+ in infrastructure sitting idle because cooling can't keep up
- **Thermal runaway events** increasing 340% since 2023
- **Climate stress:** Extreme heat waves making outdoor air cooling unreliable

The numbers are staggering:

Metric	Current State	The Problem
Global data center market	$350B by 2028	Capacity constrained by thermal limits
Cooling costs	$30-50B annually	40% of operating expenses
AI training runs	10-100 MW each	Thermal density 10x traditional compute
PUE (Power Usage Effectiveness)	1.3-1.6 average	Every 0.1 improvement = $billions saved
Unplanned downtime cost	$9,000/minute	Thermal events = catastrophic losses
Stranded GPU capacity	$50B+	Too hot to deploy at full utilization

The existing solutions are failing:

- **Legacy BMS (Building Management Systems):** Reactive, not predictive. See problems after they happen.
- **DCIM tools:** Fragmented data, no AI, designed for 2010-era workloads.
- **Manual monitoring:** Can't track millions of sensors in real-time.
- **Static cooling policies:** One-size-fits-all rules that waste energy and miss anomalies.

The CoolGrid Solution

Predictive Thermal Intelligence for the AI Era

CoolGrid is the AI-native operating system for data center thermal management.

We deploy a mesh of intelligent sensors combined with physics-informed machine learning to: 1. **Predict thermal events 30+ minutes before they occur** 2. **Optimize cooling in real-time** — rack-by-rack, second-by-second 3. **Enable 20-40% higher compute density** without thermal throttling 4. **Reduce cooling energy costs by 25-35%** 5. **Prevent catastrophic outages** before they cascade

How It Works

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                         CoolGrid Architecture



    CoolGrid            Thermal Digital        Predictive Cooling
    Sensor Mesh       Twin Engine            Orchestrator
    (Hardware)          (Physics + ML)          (Real-time Control)




    100+ sensors        CFD Simulation          HVAC/CRAC/Chiller
    per MW              + ML Prediction         Direct Integration
    Temp/Flow/          + Anomaly               + Workload Migration
    Humidity/             Detection             + Alert Routing
    Power
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Core Platform Components

1. CoolGrid Sensor Mesh

- **High-density IoT deployment:** 100+ sensors per MW of capacity
- **Multi-modal sensing:** Temperature, airflow velocity, humidity, dewpoint, pressure differential
- **Rack-level granularity:** Inlet/outlet temps for every server
- **Wireless mesh networking:** Self-healing, no cable infrastructure
- **Edge processing:** Pre-filter noise, reduce bandwidth 90%

2. Thermal Digital Twin

- **Physics-informed ML:** Combines computational fluid dynamics (CFD) with deep learning
- **Real-time simulation:** Updates every 5 seconds with live sensor data
- **What-if scenarios:** Test cooling changes before implementing
- **Predictive modeling:** 30-minute horizon with 95% accuracy
- **Anomaly detection:** Identifies thermal drift before sensors spike

3. Predictive Cooling Orchestrator

- **Direct integrations:** HVAC systems, CRAC units, chillers, liquid cooling loops
- **Workload-aware:** Receives job scheduling data to anticipate thermal loads
- **Multi-objective optimization:** Balance cooling, energy cost, and carbon intensity
- **Automated remediation:** Pre-cool zones before GPU training runs hit
- **Graceful degradation:** Coordinate workload migration before thermal throttling

4. Incident Command Center

- **Real-time 3D thermal visualization:** See hot spots in immersive view
- **Predictive alerts:** "Zone C-7 will exceed threshold in 22 minutes"

- **Root cause analysis:** Automated investigation of thermal anomalies
- **Runbook automation:** Triggered responses for common scenarios
- **Post-incident analytics:** Learn from every thermal event

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Why Now?

The Perfect Storm of 2026

1. AI Infrastructure Explosion - ChatGPT, Claude, Gemini driving unprecedented GPU deployments - NVIDIA H100/H200 clusters generating 700W+ per chip - Hyperscalers racing to build 100MW+ AI data centers - Training runs that can't be interrupted for thermal throttling

2. The AWS Wake-Up Call - May 2026 Virginia outage made every CIO rethink thermal resilience - Insurance companies now requiring thermal monitoring for coverage - Board-level attention on data center risk management

3. Sustainability Pressure - Data centers = 4% of global electricity, growing 15%/year - Cooling = 40% of that power consumption - EU Data Center Energy Efficiency Directive mandating PUE reporting - Carbon commitments forcing efficiency improvements

4. Economic Pressure - GPU scarcity = every watt of thermal headroom is revenue - Energy prices volatile, cooling costs unpredictable - Stranded capacity due to thermal limits = billions in lost opportunity

5. Technology Enablers - Edge AI chips enabling real-time sensor processing - Physics-informed ML achieving CFD-level accuracy at 1000x speed - Direct liquid cooling adoption creating new control opportunities - 5G/WiFi 6E enabling dense wireless sensor deployments

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Market Opportunity

Total Addressable Market: $50B+

Data Center Infrastructure Management (DCIM): $4.5B (2026) → $12B (2030)

Data Center Cooling Systems: $15B (2026) → $35B (2030)

Building Management Systems (Industrial): $30B+ globally

Our Beachhead: AI/HPC Data Centers

Focus on highest thermal density environments: - Hyperscaler AI training facilities (AWS, Azure, GCP) - Enterprise AI clusters (Fortune 500 GPU deployments) - Colocation providers serving AI workloads (Equinix, Digital Realty) - Sovereign AI data centers (government, defense)

Why this segment: - Thermal problems are existential, not nice-to-have - Highest willingness to pay ($1M+ ARR deals) - Reference customers drive industry adoption - Clear ROI: 1% efficiency improvement = millions saved

Expansion Path

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Year 1-2: AI/HPC Data Centers (100MW+ facilities)
     ↓
Year 3-4: Enterprise Data Centers (10-100MW)
     ↓
Year 5+: Edge Computing, Telecom, Industrial
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Business Model

Hybrid Hardware + Software Model

1. CoolGrid Sensor Kit (One-time + Subscription)

Tier	Coverage	Hardware	Annual Software	Total Y1
Starter	1 MW	$25K	$50K	$75K
Professional	5 MW	$100K	$200K	$300K
Enterprise	25 MW	$400K	$750K	$1.15M
Hyperscale	100+ MW	Custom	$2-5M	$3-8M

2. Platform Pricing (ARR per MW)

- **Core Platform:** $50K/MW/year (monitoring, alerts, visualization)
- **Predictive Module:** +$25K/MW/year (ML predictions, digital twin)
- **Orchestration Module:** +$35K/MW/year (automated control)
- **Full Stack:** $100K/MW/year

3. Value-Based Pricing Options

- **Energy savings share:** 20% of documented cooling energy reduction
- **Capacity unlock:** $500K per MW of thermal headroom recovered
- **SLA guarantees:** Premium pricing for uptime commitments

Unit Economics

Metric	Target
Hardware margin	50%+
Software margin	85%+
Blended gross margin	75%
CAC payback	12-18 months
Net retention	130%+
LTV/CAC	5x+

Competitive Landscape

Current Market Players

Company	Approach	Limitation
Schneider Electric	Legacy DCIM + cooling	Reactive, not predictive. Siloed tools.
Vertiv	Cooling hardware	No software intelligence layer
Nlyte	DCIM software	2010-era architecture, bolt-on ML
Sunbird	DCIM dashboards	Visualization only, no control
DC Brain (Google)	Internal AI cooling	Not productized, hyperscaler-only

CoolGrid Differentiation

1. AI-Native Architecture - Built from scratch for ML workloads — not retrofitted - Physics-informed models, not just pattern matching - Real-time inference at the edge

2. Predictive, Not Reactive - 30-minute prediction horizon vs. alerting on current state - Proactive cooling adjustments before problems occur - Workload-aware thermal forecasting

3. Closed-Loop Control - Direct integration with cooling systems - Automated remediation, not just recommendations - Reduces mean-time-to-resolution from hours to seconds

4. Purpose-Built Hardware - Sensors optimized for data center environments - Wireless mesh eliminates cabling complexity - Edge compute reduces latency to milliseconds

5. Hyperscaler DNA - Team built Google's internal DC cooling AI - Designed for 100MW+ scale from day one - Battle-tested on most demanding workloads

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Product Roadmap

Phase 1: Intelligent Monitoring (Months 0-12)

- CoolGrid Sensor Kit v1 (100 sensors/MW)
- Real-time thermal visualization dashboard
- Anomaly detection and alerting
- Basic digital twin (2D thermal maps)
- Integration APIs for major DCIM platforms

Milestone: 10 pilot customers, 500MW under monitoring

Phase 2: Predictive Intelligence (Months 12-24)

- Physics-informed ML prediction engine
- 3D thermal digital twin with CFD integration
- 30-minute thermal forecasting
- Workload integration (Kubernetes, Slurm, VMware)
- Energy optimization recommendations

Milestone: 50 customers, 5GW under management, $15M ARR

Phase 3: Autonomous Control (Months 24-36)

- Direct HVAC/CRAC/chiller integration
- Automated cooling orchestration
- Workload migration recommendations
- Liquid cooling support (CDU integration)
- Carbon-aware cooling optimization

Milestone: 150 customers, 25GW under management, $75M ARR

Phase 4: Platform Expansion (Months 36-48)

- Edge data center support
- Multi-site fleet management
- Capacity planning and simulation
- Marketplace for third-party integrations
- White-label offering for OEMs

Milestone: 500+ customers, $200M ARR

Go-to-Market Strategy

Phase 1: Lighthouse Customers

Target: 5-10 hyperscale/enterprise AI facilities as design partners

Approach: - Free pilot deployments (3 months) - Co-development of product features - Case study rights and references - Expedited deployment support

Ideal Customer Profile: - Operating 25MW+ of AI/HPC workloads - Experienced thermal incidents or throttling - Strong sustainability commitments - Technical team willing to engage deeply

Phase 2: Direct Enterprise Sales

Target: Fortune 500 enterprise data centers

Team Structure: - Field sales: Territory-based enterprise reps - Solutions engineering: Technical pre-sales support - Customer success: Deployment and optimization

Sales Motion: - Thermal risk assessment (free) - Pilot deployment (3 months, subsidized) - Expansion to full facility - Multi-site rollout

Phase 3: Channel Partnerships

OEM Partnerships: - Cooling system manufacturers (embed CoolGrid in units) - Server vendors (pre-integrated thermal monitoring) - DCIM platform partnerships (complement existing tools)

System Integrators: - Accenture, Deloitte (data center practices) - Specialized DC consultants

Colocation/Wholesale: - Equinix, Digital Realty, QTS partnerships - Offer to tenants as value-added service

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Financial Projections

Revenue Forecast

Year	Customers	MW Managed	ARR	Growth
2027	15	750	$5M	—
2028	60	4,000	$25M	400%
2029	175	15,000	$85M	240%
2030	400	45,000	$225M	165%
2031	800	100,000	$500M	122%

Key Assumptions

- **ACV growth:** $75K → $125K as product matures
- **Expansion revenue:** 130% net retention
- **Churn:** <5% annually (infrastructure is sticky)
- **Land-and-expand:** 80% of customers expand within 18 months

Path to Profitability

Metric	2027	2028	2029	2030
Revenue	$5M	$25M	$85M	$225M
Gross Margin	65%	72%	76%	78%

Metric	2027	2028	2029	2030
R&D	50%	40%	30%	22%
S&M	45%	40%	35%	28%
G&A	20%	15%	12%	10%
EBITDA Margin	-50%	-23%	-1%	18%

Team

Leadership (To Build)

CEO — Data Center Industry Veteran - 15+ years in data center operations or infrastructure - Experience at hyperscaler or major colocation provider - Strong enterprise sales network - Startup experience preferred

CTO — ML + IoT Expert - Built production ML systems at scale - Deep expertise in sensor networks and edge computing - PhD in thermal modeling, controls, or related field - Published research in predictive maintenance

VP Engineering — Hardware + Software - Built IoT products from prototype to scale - Experience with industrial sensor systems - Strong manufacturing and supply chain background - Data center hardware experience

VP Sales — Enterprise Infrastructure - Sold $1M+ deals to data center operators - Network in hyperscale and enterprise DC - Track record of building sales teams - Technical credibility with facilities teams

Key Hires (Year 1)

1. **ML Research Lead** — Physics-informed modeling
2. **Hardware Engineering Lead** — Sensor design and manufacturing
3. **Platform Engineering Lead** — Cloud backend and edge
4. **Solutions Architect** — Customer deployments
5. **Customer Success Lead** — Ensure time-to-value

Advisory Board

- Former Google Data Center VP (cooling AI experience)
- ASHRAE Fellow (HVAC standards expert)
- Ex-Equinix/Digital Realty CTO
- Climate tech investor

Funding Requirements

Seed Round: $5M

Use of Funds: - Core team (8-10 engineers): $2.5M - Sensor prototype and pilot hardware: $1M - Cloud infrastructure and ML development: $800K - GTM and operations: $500K - G&A buffer: $200K

Milestones: - MVP sensor kit and software platform - 5 pilot customers deployed - First predictive models validated - Series A readiness

Series A: $20M (Month 15)

Use of Funds: - Scale engineering (30+ headcount): $10M - Manufacturing scale-up: $4M - Sales and marketing expansion: $4M - Operations: $2M

Milestones: - 50 customers, $15M ARR - Autonomous control product launch - Break-even unit economics proven

Series B: $60M (Month 30)

Use of Funds: - Global expansion - Platform development - Channel partnerships - Path to profitability

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Risk Factors and Mitigations

Risk	Impact	Mitigation
Hardware manufacturing delays	Slow deployments	Multiple CM partnerships, modular design
Long enterprise sales cycles	Slow revenue growth	Freemium pilot model, strong references
Hyperscaler builds in-house	Lost market segment	Focus on enterprise, provide integration layer
Integration complexity	High implementation cost	Pre-built connectors, professional services
Competitive response	Price pressure	Patent portfolio, rapid innovation
Macroeconomic downturn	Delayed purchases	Position as cost-saving tool

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Why CoolGrid Wins

The Confluence of Forces

1. **Existential Problem:** AWS outage proved thermal management is now board-level issue
2. **Market Timing:** AI infrastructure boom creating unprecedented thermal challenges
3. **Technology Moat:** Physics-informed ML + purpose-built hardware
4. **Clear ROI:** 25% energy savings + 40% density increase = massive value
5. **Sticky Product:** Deeply embedded in operations, high switching costs
6. **Platform Potential:** Expand from thermal to full DC operations

The Vision

In 5 years, no data center will operate without AI-powered thermal intelligence.

Just as modern cars have engine management systems, data centers will have thermal intelligence platforms. CoolGrid will be the standard — preventing outages, maximizing efficiency, and enabling the AI infrastructure buildout the world needs.

The heat is rising. CoolGrid keeps the world's compute running.

Call to Action

We're seeking: - $5M Seed round to build MVP and validate with pilots - Design partner customers (25MW+ AI facilities) - Advisors with data center operations experience

Contact: - Email: founders@coolgrid.io - Web: https://coolgrid.io

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"In the AI era, the companies that master thermal intelligence will master compute. CoolGrid gives you that mastery."